Titan Hydrocarbon Protocell Framework

Polymer-Based Compartmental Evolution Under Titan Surface Conditions

Overview: Framework for modeling protocell-like structures on Titan based on polymer accumulation, phase separation, and surface-catalyzed chemistry.

Physical Environments: Atmosphere/aerosols (UV-driven chemistry), liquid methane/ethane (slow kinetics), surface/evaporites (catalytic concentration).

Protocell Definition: Polymer-rich aggregates capable of retaining reactive species and sustaining internal chemistry.

Reaction Rate Model: $k_{eff} = k0 * f_T(T) * f_{solvent} * f_{surface} * f_{cat}$

Temperature Factor: $f_T(T) = \exp[-Ea/R * (1/T - 1/T_{ref})]$

Solvent Effects: Methane-rich favors small molecules; ethane-rich stabilizes larger species; surfaces enhance concentration.

Catalysis: Tholins, minerals, and surfaces enhance reaction rates and selectivity.

Polymerization Mechanisms: Hydrocarbon growth, HCN polymerization, nitrile networks, surface-mediated reactions.

Protocell Formation: Occurs when polymer mass exceeds aggregation threshold.

Evolution: Selection based on survival, catalytic efficiency, solute retention, and stability.

Dynamics: Growth, fragmentation, fusion, exchange, catalysis.

Key Insight: Evolution driven by persistence and catalysis, not metabolism.

Interpretation: Nonaqueous evolving chemical systems, not biological life.

Next Steps: Integrate reaction networks, catalysts, aggregation, stochastic dynamics.